Assignment 2: Derivatives Pricing & Interest Rate Risk

Weight: 20% of Final Grade

Instructions:

- This assignment will be completed in **groups of 5 students**, as defined by the signup on Quercus. Submit one report with all names listed.
- Submit via Quercus: (1) your written report as a **PDF** (likely to Crowdmark, linked on Quercus), and (2) all code in a separate file or compressed folder. Only one group member should upload the submission.
- There is no page limit, but be concise. A good report is thorough but not padded.

Marking:

- **75%** — Methods and results (correct implementation, complete answers to all parts, appropriate choice of methods, accurate numerical output, properly formatted tables and figures)
- **25%** — Overall quality (clear and professional writing, thoughtful interpretation of results, demonstrated understanding of the underlying concepts, logical flow and narrative structure)

Writing Expectations: Your report should be **code-free**. The deliverable is a professional document containing writing, tables, charts, and figures—no code cells, no code snippets, no screenshots of code. Coding is a tool you use to produce results; interpretation is what you are being evaluated on. Use section headers to organize your work by problem, and write in complete sentences and paragraphs. A reader should be able to follow your entire analysis without seeing a single line of code.

Submit your code separately as a `.py` file (or Jupyter notebook) alongside your report.

You may use AI coding assistants (ChatGPT, Copilot, Claude, etc.) to help write code, but you must be able to explain every step of your analysis. You are ultimately responsible for your own work. **If you use an AI tool, you must disclose this in a note at the end of your report. Mention which tool you used, which tasks you asked it to complete, and discuss your (dis)satisfaction with its assistance.**

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Data: You are provided with a single CSV file containing option chain data from OptionMetrics for the following **five stocks**, all from the same trading date:

Ticker	Company	Sector
NVDA	NVIDIA	Technology / AI
JPM	JPMorgan Chase	Banking
XOM	ExxonMobil	Energy
LLY	Eli Lilly	Pharmaceuticals
UAL	United Airlines	Airlines

The dataset contains the following variables:

Variable	Description
`ticker`	Stock ticker symbol
`date`	Trading date (same for all rows)
`expiration_date`	Option expiration date
`cp_flag`	`C` for call, `P` for put
`strike_price`	Strike price
`best_bid`	Best bid price for the option
`best_offer`	Best offer (ask) price for the option
`open_interest`	Number of outstanding contracts
`impl_volatility`	OptionMetrics implied volatility (for reference)
`underlying_close`	Underlying stock closing price on that date

Filter the data: keep only **call options** with positive open interest, bid–ask spreads that are not excessively wide, and time to expiration between 7 days and 2 years. Use the midpoint of the bid and ask as the option price.

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1 Equity Options & the Volatility Surface

The Black-Scholes model assumes constant volatility, yet option markets price different strikes and maturities at different implied volatilities. In this section, you will extract the volatility surface from real option data and explore its structure.

Risk-free rate: You will need a risk-free rate r for the Black-Scholes formula. Find an appropriate rate from a public source (e.g., U.S. Treasury yields from FRED or the Treasury website), cite the source and the specific rate you use, and briefly justify your choice. A single flat rate is fine for this assignment.

Tasks:

(a) The Black-Scholes price for a European call is:

$$C = S\,N(d_1) - K\,e^{-rT}\,N(d_2)$$

where:

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)\,T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

and $N(\cdot)$ is the standard normal CDF. The **implied volatility** σ_{imp} is the value of σ that makes the model price equal the observed market price. Implement a root-finding procedure (e.g., Brent's method via `scipy.optimize.brentq`) to extract σ_{imp} for each call option in your dataset. Report how many options fail to produce a valid implied volatility and explain why.

(b) For each of the five stocks, plot the **volatility smile**: implied volatility on the vertical axis vs. moneyness (K/S) on the horizontal axis. Show separate curves for at least 4 different expiration dates on the same plot. Overlay a fitted curve (e.g., a quadratic in moneyness) on each expiration to highlight the shape of the smile. Describe how the shape changes with maturity.

(c) For each of the five stocks, plot the full **implied volatility surface** in 3D: moneyness on one axis, time to expiration on the other, and implied volatility on the vertical axis. You may use a scatter plot, mesh, or heatmap—whichever communicates the structure most clearly.

(d) Compare the volatility surfaces across the five stocks. The **volatility skew** measures how implied volatility varies with moneyness for a fixed expiration—specifically, the tendency for out-of-the-money puts (equivalently, low-strike calls) to have higher implied volatility than at-the-money options. A steeper skew means a larger gap between low-strike and high-strike implied volatilities. Which stock has the steepest skew? Which has the highest overall level of implied volatility? Relate these differences to the nature of each company's business and risk profile. For instance, why might a bank stock exhibit a different skew pattern than a tech stock or an airline?

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2 Option Pricing: Three Methods

In practice, derivatives are priced using a variety of techniques. In this section, you will price the same European call option using three different approaches—Black-Scholes (closed-form), Monte Carlo simulation, and a binomial lattice—and compare them.

Setup: Choose **one** of the five stocks from Problem 1. Look up and cite its most-current stock price S_0, a risk-free rate r, and the at-the-money implied volatility from your volatility surface as σ. Price an **at-the-money European call** with 6 months to expiry throughout this section.

2.1 Black-Scholes

(a) Compute the Black-Scholes price using the formula from Problem 1. This is your benchmark.

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2.2 Monte Carlo

The fundamental pricing equation in derivatives says that the price of any contingent claim is the **expected present value of its future cash flows** under the risk-neutral measure:

$$C = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\text{Payoff}(S_T)]$$

The problem is that this expectation is usually impossible to compute analytically—we don't have a closed-form expression for it. Monte Carlo sidesteps this by **brute-forcing the expectation with a sample average**: simulate N independent draws of S_T, compute the payoff for each, average them, and discount. By the law of large numbers, the sample average converges to the true expectation as $N \to \infty$. The key ingredient is a model for how the stock price evolves.

Under the risk-neutral measure, the stock price follows **geometric Brownian motion**:

$$dS_t = r\,S_t\,dt + \sigma\,S_t\,dW_t$$

This is a continuous-time stochastic differential equation (SDE). To simulate it on a computer, we need to discretize time. The **Euler-Maruyama** scheme approximates the SDE over a small time step Δt:

$$S_{t+\Delta t} = S_t + r\,S_t\,\Delta t + \sigma\,S_t\,\sqrt{\Delta t}\,Z_t, \qquad Z_t \sim \mathcal{N}(0,1)$$

The $\sqrt{\Delta t}$ appears because Brownian motion increments scale as $\Delta W \sim \mathcal{N}(0, \Delta t)$, i.e., the standard deviation of a Brownian step over time Δt is $\sqrt{\Delta t}$, not Δt. This square-root scaling is fundamental to stochastic calculus and is what distinguishes it from ordinary calculus.

Simulate daily price paths using $n = 126$ time steps ($\Delta t = T/n$) to reach the terminal price S_T.

The Monte Carlo price of a European call is then:

$$\hat{C} = e^{-rT} \cdot \frac{1}{N}\sum_{i=1}^{N} \max(S_T^{(i)} - K,\ 0)$$

where $S_T^{(1)}, \ldots, S_T^{(N)}$ are N independent simulated terminal prices. The standard error of the estimate is:

$$\text{SE} = \frac{\hat{\sigma}_{\text{payoff}}}{\sqrt{N}}, \qquad \text{where } \hat{\sigma}_{\text{payoff}} = \text{std}\left(\left\{e^{-rT}\max(S_T^{(i)} - K,\ 0)\right\}_{i=1}^{N}\right)$$

This tells you how precisely you have estimated the price. A 95% confidence interval is $\hat{C} \pm 1.96 \cdot \text{SE}$.

(a) Create a **convergence plot**: run the simulation for a range of path counts (e.g., $N = 100$ up to $N = 500{,}000$) and plot the MC price ± 2 standard errors against the BS price. At what N does the MC estimate stabilize to within 1% of the BS price? Report the MC price, standard error, and 95% confidence interval at each N. Does the MC price converge exactly to the BS price?

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2.3 Binomial Lattice

The Cox-Ross-Rubinstein (CRR) binomial tree models the stock price as moving up or down at each discrete time step. Over N steps of size $\Delta t = T/N$:

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}, \qquad q = \frac{e^{r\,\Delta t} - d}{u - d}$$

where u and d are the up and down factors, and q is the risk-neutral probability of an up move. The tree is **recombining**: an up followed by a down gives the same price as a down followed by an up, so at time step i there are only $i+1$ distinct nodes rather than 2^i. At node (i, j)—time step i with j up moves—the stock price is $S_0\, u^j\, d^{(i-j)}$.

To price the option, you work *backwards* through the tree. The logic is as follows. At expiration, the option's value is known—it is simply the payoff $\max(S_T - K, 0)$. Now step back one period. At any node one step before expiry, the stock will either go up or down next period, and in each case you know what the option will be worth (you just computed those terminal values). So the option at this node is equivalent to a one-period claim on those two known future values. Its fair price is the discounted risk-neutral expectation:

$$C_{i,j} = e^{-r\,\Delta t}\left[q\, C_{i+1,\,j+1} + (1-q)\, C_{i+1,\,j}\right]$$

This argument applies at *every* node, not just the ones adjacent to expiry. Once you have filled in the option values at step $i+1$, those become the "payoffs" for step i—the prices at which the option could be sold next period. You collapse the tree one layer at a time until you arrive at C_0, the price today.

(a) For $N = 3$ steps, **draw the full binomial tree**. At every node, show both the stock price and the option value. Walk through the backward induction explicitly—show the calculation at each node. Verify that C_0 is close to the Black-Scholes price (it won't match exactly with only 3 steps, but it should be in the right ballpark).

(b) Implement the CRR binomial tree as a general pricer. Create a **convergence plot**: choose a range of step counts from small to large, and plot the binomial price as a function of N with the Black-Scholes price as a horizontal line. Does this visualize that Black-Scholes is the continuous-time limit of the binomial model?

(c) Compare the three methods in a summary table. For each method, report the option price and the wall-clock computation time (use Python's `time` module or similar). Which method is fastest? Which is most flexible? When would you choose each one?

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3 Pricing Exotic Options

The power of Monte Carlo is that it can price *any* derivative whose payoff can be simulated, including path-dependent options where no closed-form solution exists. In this section, you will price two exotic options using Monte Carlo.

Setup: Use the same stock, parameters, and option maturity as Problem 2. For path-dependent options, simulate daily price paths using $n = 126$ time steps ($\Delta t = T/n$):

For each exotic option below, choose N large enough that the standard error of each estimate is less than 0.1% of S_0. Report the N you use and verify that this criterion is met.

Tasks:

(a) Price an **arithmetic Asian call option**. Instead of using the final stock price, the Asian call replaces S_T with the average price over the life of the option. The payoff is:

$$\max\big(\bar{S} - K,\ 0\big)\,, \qquad \text{where } \bar{S} = \frac{1}{n}\sum_{i=1}^{n} S_{t_i}$$

There is no closed-form solution for this option. Report the MC price and its standard error. How does it compare to the vanilla European call price? Explain intuitively why the averaging changes the price.

(b) Price a **floating-strike lookback call option**. A vanilla call has a fixed strike K—but what if you could set the strike equal to the lowest price the stock reached over the life of the option? That is exactly what a lookback call does. The payoff is:

$$S_T - \min_{0 \le t \le T} S_t$$

Report the MC price and its standard error. Why is the lookback option more expensive than the vanilla call? What real-world trading motivation might justify paying this premium?

(c) Build a comparison table: vanilla European call (BS), Asian call (MC), and lookback call (MC). For each, report the price and briefly explain how the payoff structure drives the price difference. Which of these three options provides the most "insurance" to the holder? Which provides the least?

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4 The CIR Model & Bond Pricing

The Cox-Ingersoll-Ross (CIR) model captures mean-reverting interest rates with volatility that scales with the level of rates, preventing negative rates:

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dW_t$$

where κ is the speed of mean reversion, θ is the long-run mean rate, and σ controls volatility. The $\sqrt{r_t}$ in the diffusion term means that volatility vanishes as rates approach zero, which prevents the continuous-time process from ever going negative. The condition $2\kappa\theta > \sigma^2$ (the **Feller condition**) makes this guarantee strict: rates stay *strictly* positive, never even touching zero.

A caveat on simulation: the Euler-Maruyama discretization does not respect this boundary exactly, since discrete steps can overshoot into negative territory. This is a numerical artifact, not a property of the model. A simple fix is to set any negative simulated rate to zero ("absorption at zero").

Parameters: Use the following calibrated parameters throughout this section:

$$\kappa = 0.2, \qquad \theta = 0.04, \qquad \sigma = 0.10$$

The initial short rate is $r_0 = 0.045$, i.e., the interest rate at time 0 is 4.5%. These parameters roughly correspond to a world where the short rate reverts slowly to 4% with moderate volatility, starting slightly above its long-run mean.

Simulation: To simulate the CIR process, use the Euler-Maruyama scheme with daily time steps ($\Delta t = 1/252$):

$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\,\Delta t + \sigma\sqrt{r_t}\,\sqrt{\Delta t}\,Z_t, \qquad Z_t \sim \mathcal{N}(0,1)$$

Bond pricing: If the rate were constant, the discount factor would simply be e^{-rT}. When the rate is stochastic, you must compound all the instantaneous rates along each path (the integral $\int_0^T r_t\,dt$, since continuously compounded interest rates compound additively) and then average the resulting discount factor across all possible paths with the expectation operator $\mathbb{E}[\cdot]$. Putting this all together, the price of a **zero-coupon bond** paying \$1 at maturity T is:

$$P(T) = \mathbb{E}\left[\exp\left(-\int_0^T r_t\,dt\right)\right]$$

Approximate this via Monte Carlo: for each simulated path, compute the discrete integral $\sum_i r_{t_i}\,\Delta t$ and take the average of $e^{-\sum r_{t_i}\Delta t}$ across all paths.

Tasks:

(a) Use Monte Carlo to price zero-coupon bonds with face value $F = \$1$ for maturities $T \in \{1, 2, 3, 5, 7, 10\}$ years. Choose N large enough that bond price standard errors are less than \$0.001 per dollar of face value. Plot 10–20 sample paths, the mean path, and the 5th/95th percentile bands. Does the mean path converge to θ? How quickly?

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(b) From the bond prices, extract the **continuously compounded yield** for each maturity. If the rate were constant at some value y, the bond price would be $P = e^{-yT}$. Inverting this:

$$y(T) = -\frac{\ln P(T)}{T}$$

This defines the yield as the single constant rate that reproduces the bond price. Even though rates are stochastic in the CIR model, $y(T)$ summarizes the average discounting over horizon T in a single number. Compute $y(T)$ for each maturity and plot the **yield curve**. Is the curve upward- or downward-sloping? Why?

(c) The CIR model has a **closed-form** bond pricing formula. For a zero-coupon bond with face value F:

$$P(T) = F \cdot A(T)\, e^{-B(T)\, r_0}$$

where, with $\gamma = \sqrt{\kappa^2 + 2\sigma^2}$:

$$B(T) = \frac{2(e^{\gamma T} - 1)}{(\gamma + \kappa)(e^{\gamma T} - 1) + 2\gamma}$$

$$A(T) = \left[\frac{2\gamma\, e^{(\kappa+\gamma)T/2}}{(\gamma + \kappa)(e^{\gamma T} - 1) + 2\gamma}\right]^{2\kappa\theta/\sigma^2}$$

Compute the closed-form prices for the same maturities and compare to your Monte Carlo estimates. Report the pricing errors. Are they within the Monte Carlo standard error?

(d) Compute the **duration** and **convexity** of a 10-year zero-coupon bond numerically.

Up to now, we have written the bond price as $P(T)$—a function of maturity. But the bond price also depends on the initial short rate r_0, and for this part we need to make that dependence explicit. Write $P(r_0)$ for the price of the 10-year zero as a function of the initial rate.

Duration measures how sensitive the bond price is to changes in rates. Formally, it is the (negative, normalized) first derivative:

$$D = -\frac{1}{P}\frac{dP}{dr_0}$$

We can approximate this using a **finite difference**: reprice the bond at $r_0 + \Delta r$ and $r_0 - \Delta r$ (use $\Delta r = 0.001$, i.e., 10 basis points) using the closed-form CIR bond pricing formula from part (c), and apply the central difference formula:

$$\frac{dP}{dr_0} \approx \frac{P(r_0 + \Delta r) - P(r_0 - \Delta r)}{2\,\Delta r}$$

so that:

$$D \approx -\frac{1}{P} \cdot \frac{P(r_0 + \Delta r) - P(r_0 - \Delta r)}{2\,\Delta r}$$

Convexity is the normalized second derivative, which captures the curvature of the price–rate relationship. The analogous finite difference approximation is:

$$C \approx \frac{1}{P} \cdot \frac{P(r_0 + \Delta r) - 2P(r_0) + P(r_0 - \Delta r)}{(\Delta r)^2}$$

Report both. How does the CIR duration of a 10-year zero compare to the simple approximation that duration $\approx$ maturity for a zero-coupon bond? Implement duration and convexity as reusable functions—you will need them in Problem 5.

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5 Hedging a Bank's Interest Rate Risk

A Canadian bank holds a portfolio of long-duration government bonds on its balance sheet. When interest rates rise, the market value of these assets falls—precisely the risk that brought down Silicon Valley Bank in 2023. In this section, you will quantify that risk and evaluate a hedging strategy using interest rate derivatives priced under the CIR model.

5.1 The Bank's Bond Portfolio

Setup: The bank holds the following Government of Canada bonds (semi-annual coupons):

Bond	**Face Value ($M)**	**Coupon Rate**	**Maturity**
Bond A	500	2.5%	5 years
Bond B	300	2.0%	10 years
Bond C	200	1.5%	20 years

Use the same CIR parameters from Problem 4.

Tasks:

(a) Price each bond under the CIR model using the closed-form zero-coupon formula from Problem 4. A coupon-bearing bond is just a portfolio of zero-coupon bonds—one for each cash flow. A bond with face value F, coupon rate c, and semi-annual payments has cash flows of $\frac{c}{2} \cdot F$ at each coupon date and F at maturity. Its price is:

$$V = \sum_{i=1}^{2T} \frac{c}{2} F \cdot A(t_i)\, e^{-B(t_i)\, r_0} \;+\; F \cdot A(T)\, e^{-B(T)\, r_0}$$

where $t_i = i/2$ are the semi-annual coupon dates. Report the market value of each bond and the **total portfolio value**.

(b) Compute the **portfolio duration** as the market-value-weighted average of the individual bond durations. Interpret the number: if rates rise by 100 basis points, approximately how much does the portfolio lose?

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5.2 Stress Testing

Tasks:

(a) Simulate a **rate shock**: set $r_0 = 0.045 + 0.02 = 0.065$ (a parallel 200bp increase) and reprice the entire portfolio. Report the dollar loss and the percentage loss. Compare the actual loss to the duration-based approximation. How much of the gap is explained by convexity?

(b) Now run a **Monte Carlo stress test**. Rather than a deterministic shift, simulate 10,000 CIR paths forward by 1 year, starting from $r_0 = 0.045$. Each path gives you a realized short rate r_1 at the 1-year horizon.

(c) For each simulated r_1, reprice the portfolio at the new rate. Each bond now has one fewer year to maturity (4, 9, and 19 years respectively), so use the closed-form CIR formula with $r_0 \leftarrow r_1$ and the reduced maturities. Compute the **1-year profit and loss (P&L)** for each path:

$$\text{P\&L} = \underbrace{V_1(r_1)}_{\text{end-of-year portfolio value}} + \underbrace{\text{Coupons received}}_{\text{income during the year}} - \underbrace{V_0}_{\text{beginning-of-year value}}$$

A negative P&L is a loss.

(d) Plot the **distribution of 1-year portfolio P&L** across the 10,000 scenarios. Report the mean, 5th percentile, and 1st percentile losses. These are your Value-at-Risk estimates at the 95% and 99% confidence levels.

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5.3 Hedging with an Interest Rate Cap

An **interest rate cap** pays the holder when the short rate exceeds a strike rate $\bar{r}$. It is a portfolio of **caplets**, each of which pays:

$$\text{Caplet payoff at } t_i = \delta \cdot N \cdot \max(r_{t_i} - \bar{r},\ 0)$$

where δ is the accrual period (e.g., 0.5 for semi-annual) and the payments occur at each reset date $t_1, t_2, \ldots, t_m$. The **notional** N is the reference principal on which the interest payments are computed—it determines the dollar scale of the cap's payoffs but is never itself exchanged. Think of it as a dial that controls how much protection the cap provides. The cap price is the sum of discounted expected caplet payoffs under the risk-neutral measure.

Setup: Consider a 5-year interest rate cap with:

- Notional: equal to the total portfolio face value ($1,000M)
- Semi-annual accrual period ($\delta = 0.5$, so 10 caplets)
- Strike rate: to be explored below

Tasks:

(a) For a range of strike rates $\bar{r} \in \{5.00\%, 5.25\%, 5.50\%, \ldots, 8.00\%\}$ (in 25bp increments), price the interest rate cap via Monte Carlo under the CIR model. For each simulated path, compute each caplet's payoff, discount it back to time 0 using the path's realized short rates, and average across paths.

(b) Now assume a 200bp rate shock ($r_0 \to 0.065$). For each strike, reprice the cap at the shocked rate and compute the **net P&L of the hedged portfolio**:

$$\text{Net P\&L} = \underbrace{\text{Portfolio loss}}_{\text{from 5.2(a)}} + \underbrace{\text{Cap gain}}_{\text{shocked price}-\text{original price}} - \underbrace{\text{Cap premium}}_{\text{original price}\times N}$$

Plot the net hedged P&L as a function of the strike rate. Which strike minimizes the total cost (loss after hedging plus premium paid)? Is there a strike that makes the hedge worthwhile?

(c) Interpret the plot from part (b). Discuss the trade-off between the cost of hedging (the cap premium) and the effectiveness of the hedge (the net P&L). What does the shape of the curve reveal about using options to hedge interest rate risk? What are the limitations of using a 5-year cap to hedge a portfolio that includes 20-year bonds?

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